

Loop holonomy under noise: invariants of lossy operation systems at three layers

Abstract

We replace the group-valued operations of an operation system (Γ, A, g) by lossy channels and follow the holonomy through the three layers of the stratification. For shift-covariant uniform noise of rate ε we prove the following.

- The loop operator has character eigenvalues $(1 - \varepsilon)^{|L|} \chi(h)$: the modulus records the loss and the phase records the holonomy exactly for every $\varepsilon < 1$.
- Every per-context Shannon functional, and the loop-composite information, is independent of the holonomy.
- On the edge-context empirical model of a single cycle, the contextual fraction is $\max(0, 1 - |L|\varepsilon/2)$, independent of $|A|$ and of the holonomy.

On a general graph we prove $\text{CF} \geq \max(0, 1 - \ell_{\min}\varepsilon/2)$, where $\ell_{\min}$ is the length of the shortest cycle with nonzero holonomy, and we observe equality in 44 configurations. Information functionals do not determine the contextual fraction; we give counterexamples. The three layers respond to the same noise through different invariants, and no invariant of one layer determines another.

1 Setting

Let $\Gamma = (V, E)$ be a finite connected graph and A a finite abelian group of order n . Each edge (a, b) carries a doubly stochastic matrix $K_{ab} \in \mathbb{R}^{A \times A}$, where row x is the law of the colour at b given colour x at a . Colours are uniform at every vertex, and uniformity is preserved by doubly stochastic channels. Hence the Bayesian inverse of K_{ab} is exactly $K_{ab}^\top$, and a loop may be traversed in either direction.

Definition 1 (Noisy shift). For $g \in A$ let S_g be the permutation matrix of $x \mapsto x + g$ and $U = \frac{1}{n} \mathbf{1}\mathbf{1}^\top$. The *noisy shift* of rate $\varepsilon \in [0, 1]$ is

$$K = (1 - \varepsilon) S_g + \varepsilon U.$$

More generally, a *covariant* channel is $K = C_\mu S_g$, where $C_\mu[x, y] = \mu(y - x)$ is convolution by a probability measure μ on A .

Definition 2 (Loop operator, edge-context model). For a closed walk $L = (v_0, v_1, \dots, v_{|L|} = v_0)$, the *loop operator* is $M_L = \prod_j K_{v_j v_{j+1}}$, using $K^\top$ on edges traversed against their orientation. The *edge-context model* e_K has one measurement per vertex (its colour), one context per edge, and law $\frac{1}{n} K_{ab}$ on context (a, b) .

By double stochasticity, e_K is no-signalling. At $\varepsilon = 0$ it is the all-versus-nothing model e_g of the operation system: its support is the solution set of the linear equations $c_b - c_a = g_{ab}$ over A [3]. We measure contextuality by the contextual fraction CF [4] and by the Čech obstruction γ [2], and we write $\text{NCF} = 1 - \text{CF}$. For characters $\chi \in \hat{A}$ we write $\hat{\mu}(\chi) = \sum_z \mu(z) \chi(z)$.

2 Layer 1: the loop operator

Proposition 3 (Character eigenvalues). *Let every edge of a simple cycle L , oriented along the cycle, carry a covariant channel $C_{\mu_e} S_{g_e}$, and let $h = \sum_e g_e$. Then every character χ is an eigenvector of M_L with eigenvalue*

$$\lambda_\chi = \left(\prod_{e \in L} \hat{\mu}_e(\chi) \right) \chi(h).$$

For the noisy shift, $M_L = (1 - \varepsilon)^{|L|} S_h + [1 - (1 - \varepsilon)^{|L|}] U$, so $\lambda_{\mathbf{1}} = 1$ and $\lambda_\chi = (1 - \varepsilon)^{|L|} \chi(h)$ for $\chi \neq \mathbf{1}$.

Proof. First, $(S_g \chi)(x) = \chi(x + g) = \chi(g) \chi(x)$. Second, $(C_\mu \chi)(x) = \sum_y \mu(y - x) \chi(y) = \hat{\mu}(\chi) \chi(x)$. All factors are therefore diagonal in the character basis, so they commute and the eigenvalues multiply. For the noisy shift, $\hat{\mu}(\chi) = 1 - \varepsilon$ for $\chi \neq \mathbf{1}$, and $S_g U = U S_g = U$. $\square$

Corollary 4. *The labelled eigenvalue λ_χ is independent of the base point of the loop. If every $\hat{\mu}_e(\chi)$ is real and positive (for instance, symmetric noise with nonvanishing Fourier coefficients), then $\arg \lambda_\chi = \arg \chi(h)$. For the noisy shift, h is recovered exactly for all $\varepsilon < 1$.*

Remark 5 (What the plain spectrum sees). The multiset of eigenvalues of M_L is base-point free for any channels, since AB and BA share their nonzero spectrum. It does not determine h . For the noisy shift on $\mathbb{Z}/n$ it equals $\{1\} \cup \{(1 - \varepsilon)^{|L|} e^{2\pi i k h/n} : k \neq 0\}$, which depends on h only through its order; on $\mathbb{Z}/5$ every $h \neq 0$ gives the same multiset. For asymmetric noise the phase is contaminated. With $\mu = (0.7, 0.2, 0, 0, 0.1)$ on $\mathbb{Z}/5$ and $h = 2$, the readout is 2.285, offset by the noise-only phase 0.285.

3 Layer 2: information is blind to the holonomy

Proposition 6. *Let F be any functional of a context law that is invariant under relabelling the outcomes of each variable; every Shannon functional is of this kind. Then F takes the same value on $\frac{1}{n} N S_g$ for every g and every doubly stochastic N . In particular, per-edge information never depends on g , whether the channel is invertible or lossy.*

Proof. Right multiplication by S_g relabels the values of the second variable. $\square$

Proposition 7 (Closed forms). *For the noisy shift and uniform input,*

$$I(c_a; c_b) = \log_2 n + \left(1 - \varepsilon + \frac{\varepsilon}{n}\right) \log_2 \left(1 - \varepsilon + \frac{\varepsilon}{n}\right) + (n - 1) \frac{\varepsilon}{n} \log_2 \frac{\varepsilon}{n}.$$

The loop-composite information $I(X; M_L X)$ is given by the same expression with ε replaced by $\varepsilon_L = 1 - (1 - \varepsilon)^{|L|}$. Both are independent of h .

Proof. Each row of K puts mass $1 - \varepsilon + \varepsilon/n$ on one colour and ε/n on each of the other $n - 1$. The loop case follows from Proposition 3. $\square$

ε	I edge	I loop, $ L = 3$	$ \lambda_\chi $, $ L = 3$	CF, $\ell_{\min} = 3$
0	2.3219	2.3219	1.0000	1.000
0.05	1.9996	1.5816	0.8574	0.925
0.10	1.7597	1.1340	0.7290	0.850
0.20	1.3676	0.5761	0.5120	0.700
0.30	1.0469	0.2726	0.3430	0.550
0.50	0.5510	0.0406	0.1250	0.250

Table 1: Noisy shift on $\mathbb{Z}/5$ (information in bits). The information columns are identical for flat and non-flat systems.

4 Layer 3: the contextual fraction

For a colouring $c : V \rightarrow A$ and an edge $e = (a, b)$, write $d_e(c) = c(b) - c(a) - g_e$ for its *discrepancy*. Around a cycle L oriented along its edges, $\sum_{e \in L} d_e(c) = -h_L$.

The cells of context e split into discrepancy classes. The class $d = 0$ has total mass $1 - \varepsilon + \varepsilon/n$, and each class $d \neq 0$ has total mass ε/n , spread evenly over its n cells.

Theorem 8 (Single cycle). *Let Γ be a cycle of length $L \geq 3$ with noisy shifts of rate ε and holonomy $h \neq 0$ in a finite abelian group A . Then*

$$\text{CF}(e_K) = \max\left(0, 1 - \frac{L\varepsilon}{2}\right).$$

In particular, the threshold $\varepsilon^ = 2/L$ is independent of $|A|$ and of h .*

Proof. Upper bound on NCF. The non-contextual fraction is the value of the LP $\max \sum_c b_c$ subject to $b \geq 0$ and $\sum_{c: c|_e=s} b_c \leq e_K(e, s)$ for every edge e and section s . A feasible point of its dual is a weight $y_e(s) \geq 0$ with $\sum_e y_e(c|_e) \geq 1$ for every colouring c . Take $y_e(s) = w(d)$, where d is the discrepancy of s and

$$w(0) = 0, \quad w(-h) = 1, \quad w(d) = \frac{1}{2} \text{ otherwise.}$$

Every colouring has at least one nonzero discrepancy, since they sum to $-h \neq 0$. If it has exactly one, that one equals $-h$ and contributes 1. Otherwise at least two nonzero discrepancies contribute at least $\frac{1}{2}$ each. So the weight is feasible, and its value is

$$L \left[\frac{\varepsilon}{n} \cdot 1 + (n-2) \frac{\varepsilon}{n} \cdot \frac{1}{2} \right] = \frac{L\varepsilon}{2},$$

so $\text{NCF} \leq L\varepsilon/2$.

Lower bound on NCF (packing). Assume $L\varepsilon/2 \leq 1$. A colouring is fixed by its discrepancy vector (summing to $-h$) together with $c(v_0)$. Use two families, each spread uniformly over $c(v_0) \in A$:

- (A) for each edge k , discrepancy $-h$ on k and 0 elsewhere, with mass ε/n ;
- (B) for each ordered pair of distinct edges (k, l) and each $a \notin \{0, -h\}$, discrepancies a on k and $-h - a$ on l , with mass $\beta = \varepsilon/(2n(L-1))$.

Family (A) uses exactly the mass ε/n of class $-h$ on each edge. For $a \notin \{0, -h\}$, class a on edge k is used by the triples (k, l, a) and $(l, k, -h - a)$, which is $2(L-1)\beta = \varepsilon/n$, again exactly its mass.

Uniform spreading over $c(v_0)$ spreads each class evenly over its n cells, so every off-graph cell is filled to capacity and not beyond.

The total mass is $L\frac{\varepsilon}{n} + L(L-1)(n-2)\beta = \frac{L\varepsilon}{2}$. On each edge the on-graph cells receive $\frac{L\varepsilon}{2} - \varepsilon(1 - \frac{1}{n})$, while their capacity is $1 - \varepsilon(1 - \frac{1}{n})$. The capacity constraint is therefore equivalent to $L\varepsilon/2 \leq 1$, and it is tight exactly at $\varepsilon = 2/L$.

Beyond the threshold. The model is affine in ε . It is non-contextual at $\varepsilon = 2/L$ (just shown) and at $\varepsilon = 1$ (a product of uniform laws). Since the non-contextual set is convex, it is non-contextual on the whole interval between them. $\square$

Remark 9. For $n = 2$, Theorem 8 is the chained Bell inequality [6]. The certificate was also checked exhaustively over all colourings for $(n, L, h) \in \{(2, 3, 1), (3, 4, 2), (5, 3, 2), (5, 4, 1), (4, 3, 2)\}$. The LP agrees with the formula for $n \in \{2, 3, 5\}$, $L = 3, \dots, 6$, and for $A \in \{\mathbb{Z}/2 \times \mathbb{Z}/2, \mathbb{Z}/2 \times \mathbb{Z}/4, \mathbb{Z}/3 \times \mathbb{Z}/3\}$.

Theorem 10 (Any graph, lower bound). *For noisy shifts on any finite connected graph with nonzero class,*

$$\text{CF}(e_K) \geq \max\left(0, 1 - \frac{\ell_{\min}\varepsilon}{2}\right),$$

where $\ell_{\min}$ is the length of the shortest simple cycle with nonzero holonomy.

Proof. Apply the dual certificate of Theorem 8 on the edges of a shortest non-flat cycle, with weight 0 elsewhere. Every colouring of Γ restricts to a colouring of that cycle, whose discrepancies (signed along the cycle) sum to its nonzero holonomy. $\square$

Observation 11 (Equality, measured). Equality holds in Theorem 10 in all 44 configurations tested, for ε up to 0.8. These cover θ -graphs $\theta(2, 3, 4)$, $\theta(3, 3, 3)$, $\theta(1, 4, 4)$ and $\theta(2, 2, 5)$; K_4 ; two cycles sharing an edge; two cycles sharing a vertex; and two triangles joined by a path. Only $\ell_{\min}$ matters, not the cycle basis. In $\theta(2, 2, 5)$ with the 4-cycle flat and only the 7-cycles non-flat, the slope is $7/2$. A proof of the matching packing on general graphs is open.

Remark 12 (γ). For every $\varepsilon > 0$ the support of e_K is full, so $\gamma = 0$. At $\varepsilon = 0$, $\gamma(s) \neq 0$ for every section over every nonzero coefficient ring, because each context's support is the graph of a bijection.

5 What does not hold

Counterexample 13 (Information does not determine CF). Consider a 3-cycle and a 5-cycle, each with noisy shifts at $\varepsilon = 0.2$ over $\mathbb{Z}/3$ and holonomy 1. Every edge has the same information, yet $\text{CF} = 0.70$ and 0.50 .

Counterexample 14 (Non-covariant noise). Let $K_e = ((1 - \varepsilon)I + \varepsilon N_e)S_{g_e}$, where N_e is a random mixture of permutations. Then flat systems are already contextual, with CF up to 0.24 over 40 random noise draws at $\varepsilon \in \{0.2, 0.5\}$ on a 4-cycle over $\mathbb{Z}/5$. The phase readout of $h = 2$ ranges over $[1.1, 3.1]$. Noise and holonomy are no longer separable.

Counterexample 15 (Entropy is not a function of Fourier moduli). Take $p = a * b$ and $q = a * \tilde{b}$ with $\tilde{b}(x) = b(-x)$ on $\mathbb{Z}/7$. They have identical $|\hat{p}| = |\hat{q}|$, but $H(p) = 2.7714$ and $H(q) = 2.7689$ bits. The statement “information depends only on $|\lambda_\chi|$ ” holds for the uniform-noise family and not in general.

6 Summary

Layer	Invariant	Response to ε	Detects h until
1 Operations	labelled eigenvalue λ_χ of M_L	modulus $(1 - \varepsilon)^{ L }$, phase unchanged	$\varepsilon \rightarrow 1$
2 Probability	per-edge and loop I	decreases, independent of h	never
3 Outcomes	CF	$1 - \ell_{\min}\varepsilon/2$	$\varepsilon = 2/\ell_{\min}$
3 Outcomes	γ	support becomes full	$\varepsilon = 0$ only

Table 2: The same noise acting on three invariants.

For shift-covariant uniform noise, the modulus of the loop operator’s character eigenvalues determines every Shannon functional of the loop channel, and the phase determines h . Per-context and loop-composite information are independent of h . The contextual fraction is certified by an LP dual and is not determined by information functionals. The labelled phase detects h for every $\varepsilon < 1$, including $\varepsilon \geq 2/\ell_{\min}$, where CF = 0. The layers are therefore separated under noise, not bridged.

7 Corrections to the previous draft

1. *Attribution.* No “Brunner–Bergh–Gisin–Vertesi” framework could be identified. The Layer 3 geometry is that of the local polytope [5], the contextual fraction [4] and sheaf contextuality [1]. Vigneaux’s framework [8, 7] is information cohomology on sites where every set of observables has a joint; the contextuality statements here are not his.
2. *Information formula.* The two-point form $\log_2 n + (1 - \varepsilon) \log_2(1 - \varepsilon) + \varepsilon \log_2(\varepsilon/n)$ is replaced by Proposition 7. At $n = 5$, $\varepsilon = 0.2$ the old form gives 1.1356; the correct value is 1.3676.
3. *Conditional entropy.* The earlier expression gave $3.51 > \log_2 5$ bits, which is impossible. The correct value is $H = \log_2 n - I$.
4. “*Information bounds dictate CF*” is false (Counterexample 13).
5. “*Entropies depend on singular values or on $|\lambda|$* ” is false in general (Counterexample 15).
6. “*The spectrum reads h* ” becomes the labelled eigenvalue (Corollary 4); the multiset sees only the order of h .
7. *Spectral bridge.* The phase survives beyond CF = 0, which is a separation, not a bridge.
8. *Threshold.* $\varepsilon^* = 2/|L|$ becomes $2/\ell_{\min}$ on general graphs: proved as a bound, observed as equality.
9. *Loop joint.* The loop joint $P(X_0, X_L)$ is a law derived from composing channels, not a section of an information structure.

8 Open problems

1. Prove equality in Theorem 10, by extending the cycle packing to general graphs.
2. Find an invariant for non-covariant noise that separates the noise from the holonomy.

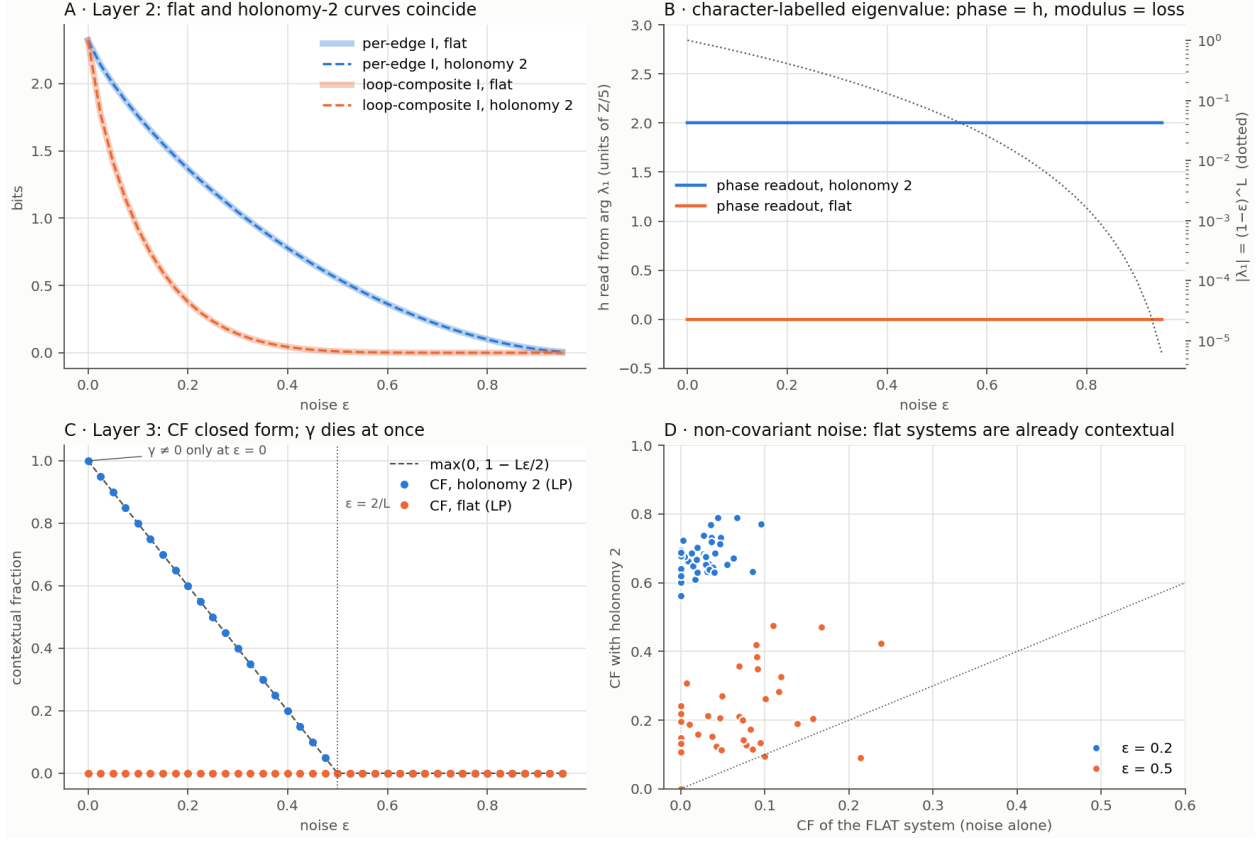


Figure 1: A 4-cycle over $\mathbb{Z}/5$ with $h = 2$, compared with a flat system. (A) Per-edge and loop information coincide for the two systems. (B) The labelled phase reads $h = 2$ while the modulus decays as $(1 - \epsilon)^4$. (C) CF follows $\max(0, 1 - L\epsilon/2)$; the flat system has CF = 0. (D) Under non-covariant noise, flat systems are already contextual.

3. Estimate λ_χ from edge readings: which observation designs make the phase identifiable, and at what sample cost? This links the present results to Problem 26.

Reproduction. All statements are checked by `tests/test_channels.py` (81 tests pass in total). The figure is produced by `examples/channels.py`. The functions are `noisy_shift_information`, `loop_information`, `cf_noisy_cycle`, `cf_noisy_graph`, `holonomy_girth` and `character_value` in `amb_vigneaux/channels.py`.

References

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